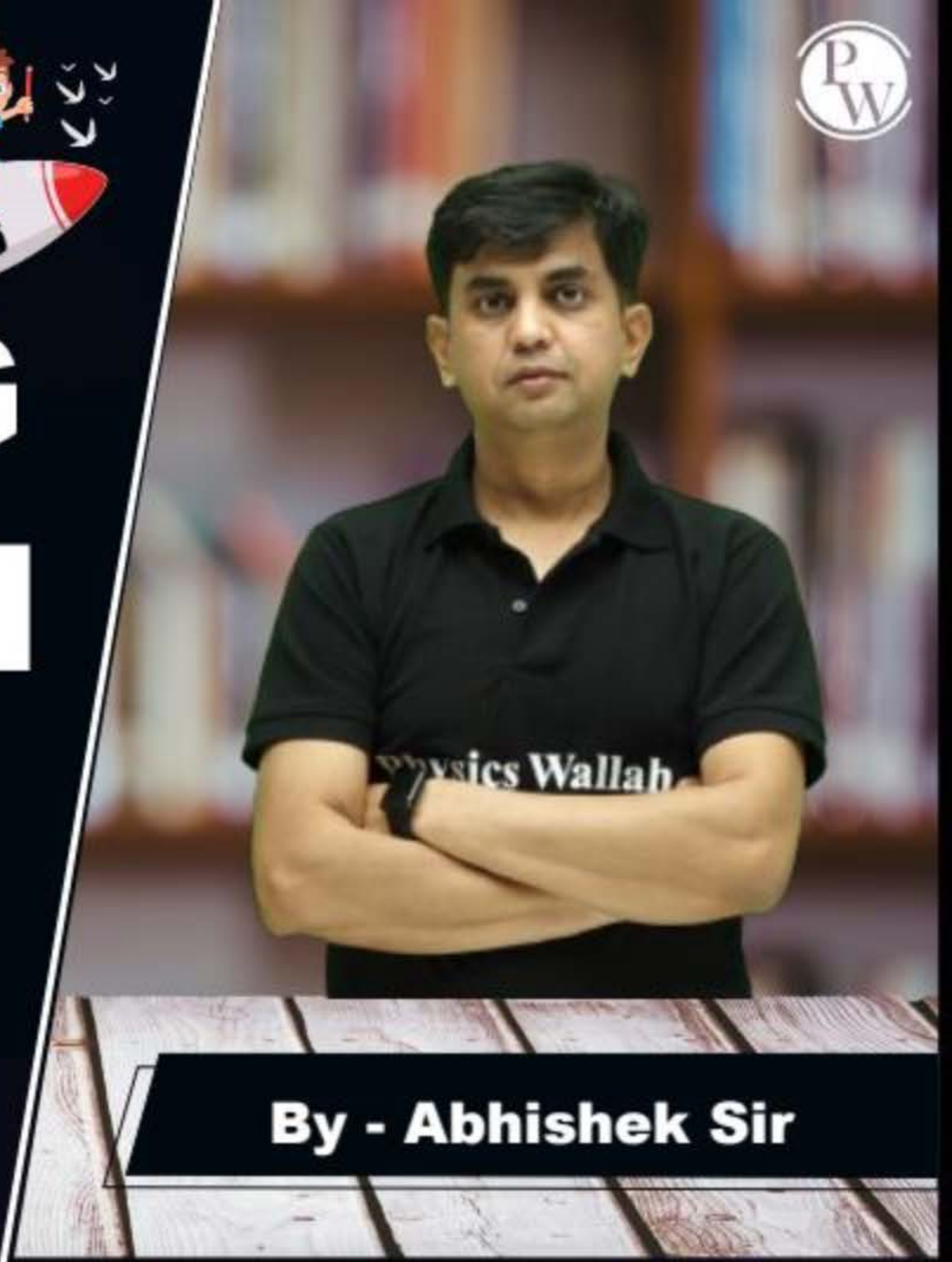
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 03



By - Abhishek Sir

Recap of Previous Lecture

Topic UDP

Topic TCP

Topics to be Covered

Topic TCP Header

Topic TCP Connections

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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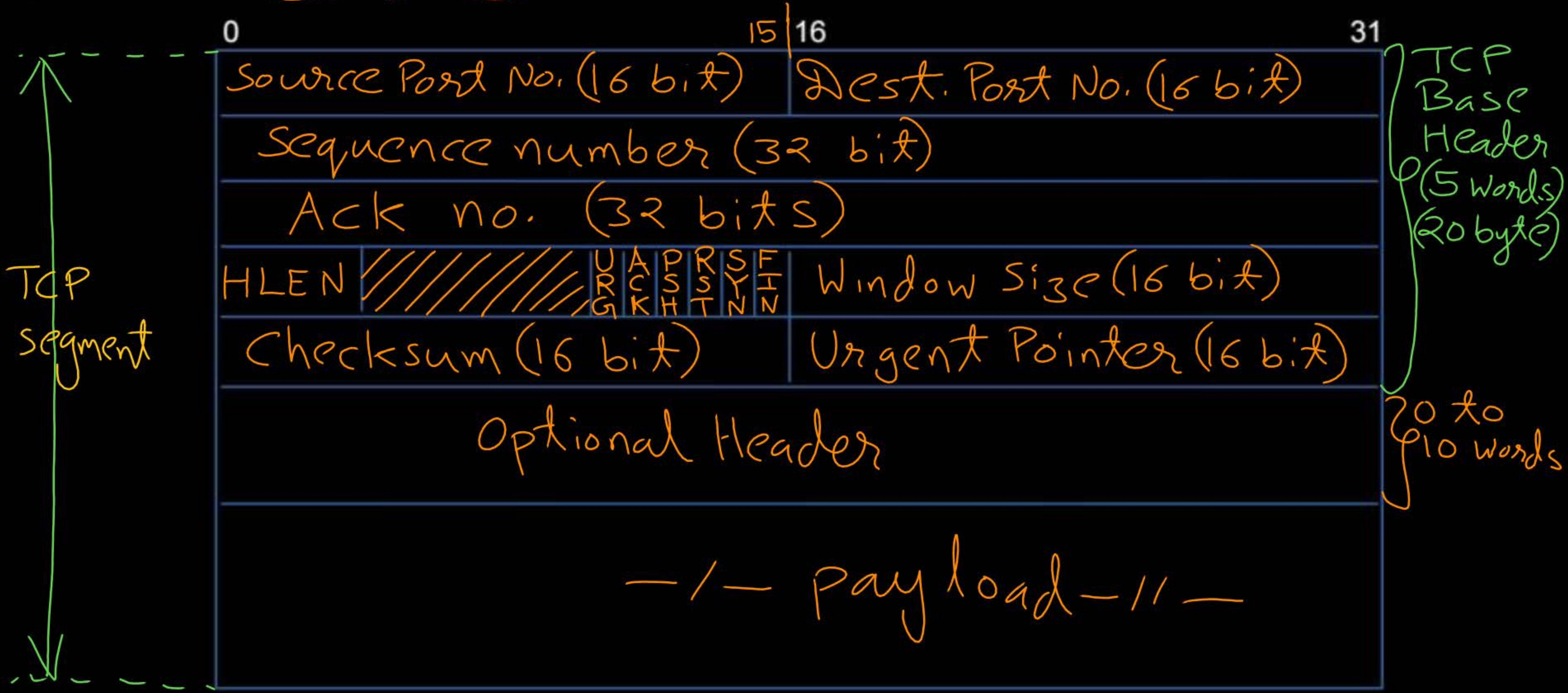
Topic : TCP Header



Topic : TCP Header

=> Source port number : 16-bits

=> Destination port number : 16-bits

=> Sequence Number : 32-bits

=> Acknowledgment Number : 32-bits

Topic : Header Length

→ Header Length [HLEN]

→ HLEN field is 4 bits long

→ Size of header in words
[Word of 4 bytes]

Topic : Header Length

→ Minimum Header Size = 5 Words (20 Bytes)

$$\boxed{5 <= \text{HLEN} <= 15}$$

→ Maximum Header Size = 15 Words (60 Bytes)

Topic : TCP Flag Bits

$\Rightarrow$ 6 flag bits

=> URG	:	Urgent
=> ACK	:	Acknowledgement
=> PSH	:	Push
=> RST	:	Reset
=> SYN	:	Synchronize
=> FIN	:	Finish

Topic : Checksum

-> Checksum : 16-bits

-> 16-bit one's complement of the one's complement sum of all 16-bit words in TCP segment including pseudo header

-> Error detection in TCP is compulsory [Unlike UDP]

Topic : IPv4 Pseudo-Header

0		16 … 31
Source IPv4 Address (32 bits)		
Destination IPv4 Address (32 bits)		
Reserved	Protocol Type	TCP Segment Length (16 bits)

Topic : TCP Header

<table>
<tr><td>0</td><td></td><td></td><td>16</td><td>31</td></tr>
<tr><td colspan="3">Source Port Number (16 bit)</td><td colspan="2">Destination Port Number (16 bit)</td></tr>
<tr><td colspan="5">Sequence Number (32 - bits)</td></tr>
<tr><td colspan="5">Acknowledgment Number (32 - bits)</td></tr>
<tr><td>HLEN</td><td></td><td>6 Flag bits</td><td colspan="2">Window Size (16 bit)</td></tr>
<tr><td colspan="3">Checksum (16 bit)</td><td colspan="2">Urgent Pointer (16 bit)</td></tr>
<tr><td colspan="5">Optional Header</td></tr>
<tr><td colspan="5">Payload</td></tr>
</table>

Topic : TCP Header

IPv4 Pseudo Header (12 bytes)

TCP Header (20 - 60 bytes)
[Including Optional Header]

Payload

Topic : TCP Operation

Three phases of TCP operation :

1. Connection establishment
   → 3-way handshake process that establishes a connection

V. Imp.

2. Data transfer ⟹ Imp.

3. Connection termination
   → 4-way handshake process that closes the connection

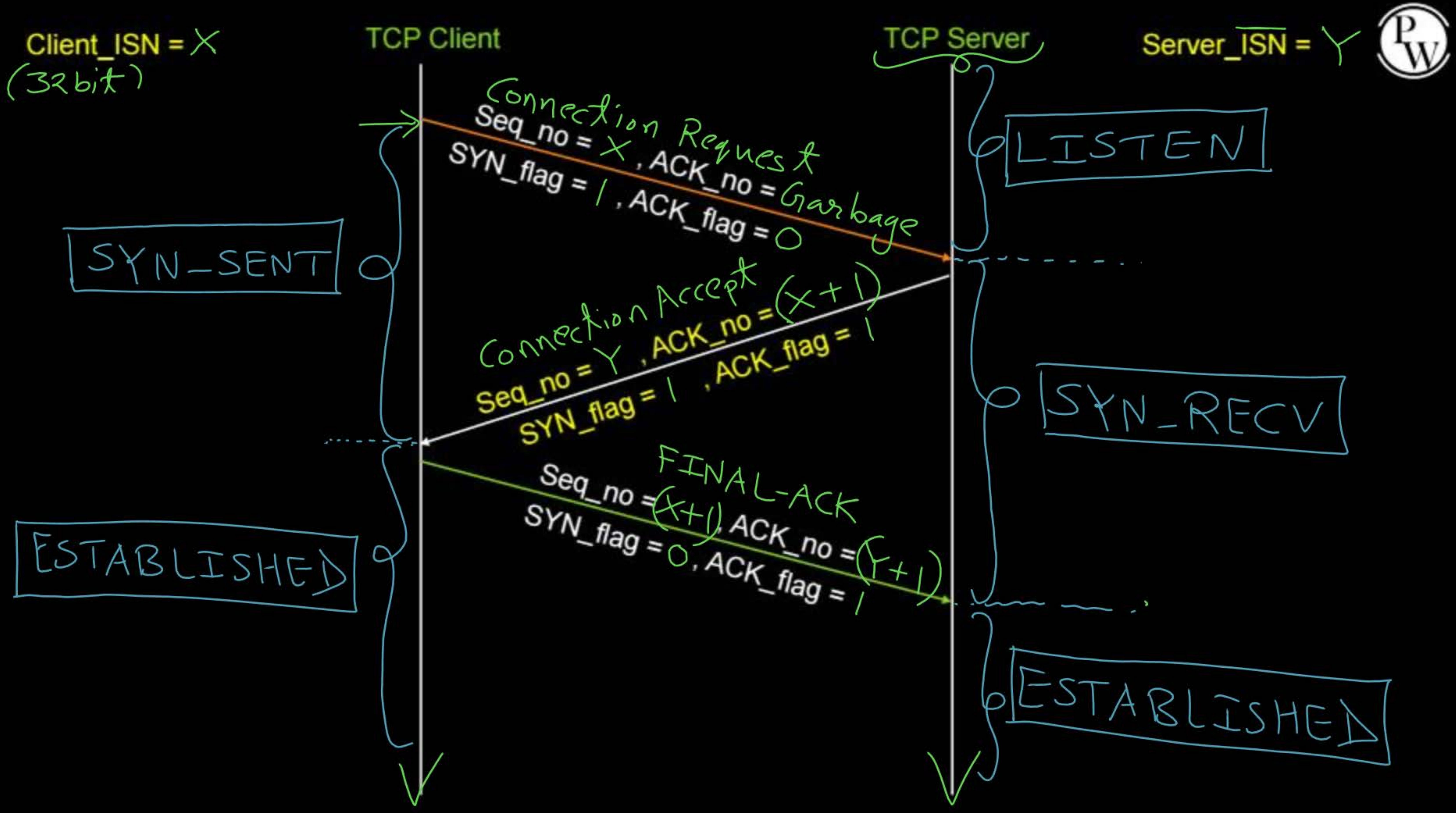
⟹ one question in gate PYQ CS-2017

Topic : TCP Connection Establishment

→ Connection establishment between TCP client and TCP server

→ 3-way handshake process

→ Always TCP client initiate the connection request to TCP server



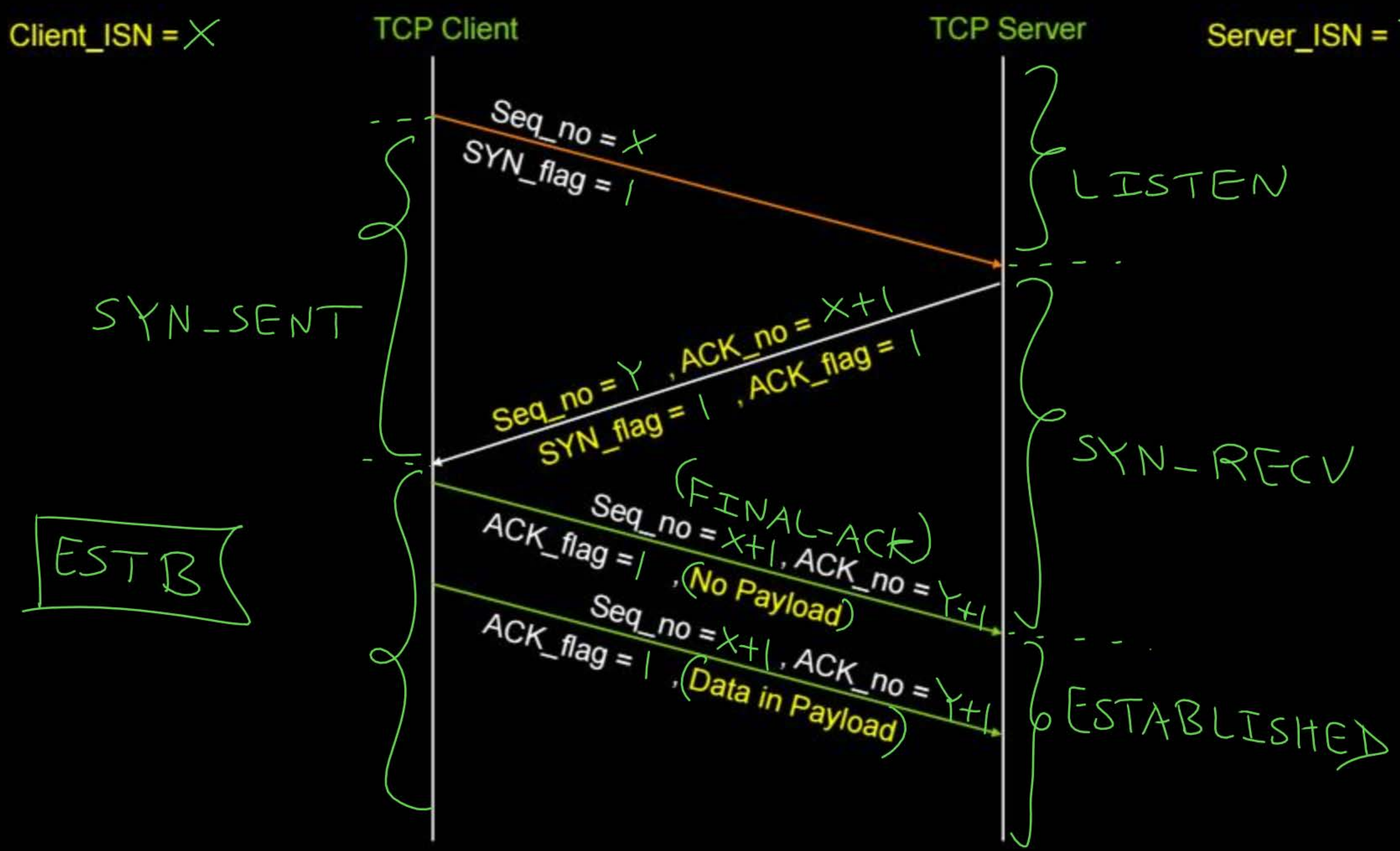
Client_ISN = X
(32bit)
TCP Client
TCP Server
Server_ISN = Y
Connection Request
Seq_no = X , ACK_no = Garbage
SYN_flag = 1 , ACK_flag = 0
LISTEN
SYN_SENT
Connection Accept
Seq_no = Y , ACK_no = (X+1)
SYN_flag = 1 , ACK_flag = 1
SYN_RECV
FINAL-ACK
Seq_no = (X+1), ACK_no = (Y+1)
SYN_flag = 0, ACK_flag = 1
ESTABLISHED
ESTABLISHED

Topic : TCP Connection Establishment

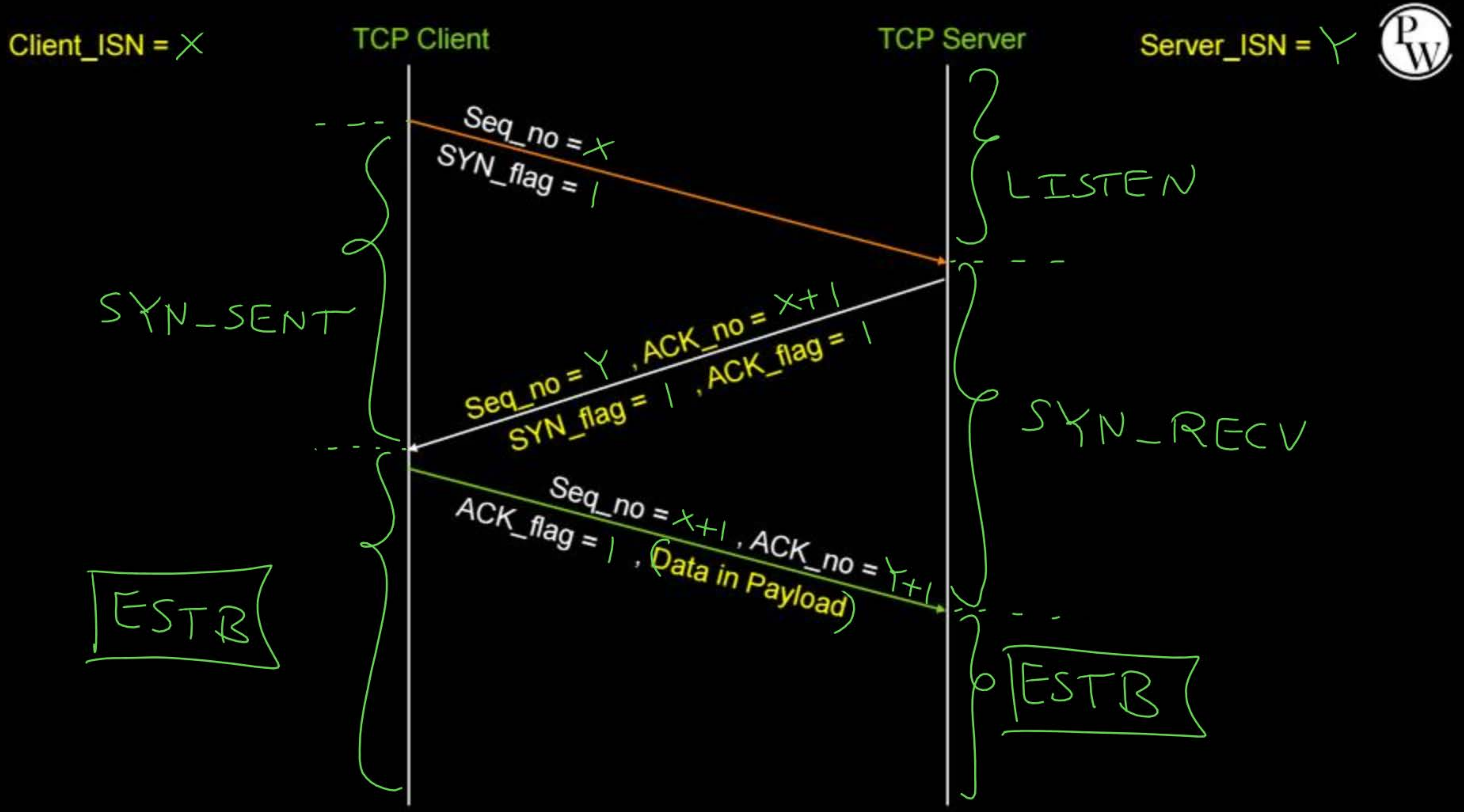
→ Initial Sequence Number : if SYN flag is on
[both TCP client and server randomly chooses their initial sequence number, to prevent from some kind of attacks]

→ SYN segment do not carry user data [no any payload]

→ SYN segment consume one sequence number



Client_ISN = X
TCP Client
TCP Server
Server_ISN = Y
Seq_no = X
SYN_flag = 1
LISTEN
SYN_SENT
Seq_no = Y , ACK_no = X+1
SYN_flag = 1 , ACK_flag = 1
SYN_RECV
(FINAL-ACK)
Seq_no = X+1, ACK_no = Y+1
ACK_flag = 1 , (No Payload)
ESTB
Seq_no = X+1 , ACK_no = Y+1
ACK_flag = 1 , (Data in Payload)
ESTABLISHED



Client_ISN = X
TCP Client
TCP Server
Server_ISN = Y
Seq_no = X
SYN_flag = 1
LISTEN
SYN_SENT
Seq_no = Y , ACK_no = X+1
SYN_flag = 1 , ACK_flag = 1
SYN_RECV
Seq_no = X+1 , ACK_no = Y+1
ACK_flag = 1 , (Data in Payload)
ESTB
ESTB

[MSQ] (H.W.) IIT-R [GATE-2025][1 Mark]

#Q. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

A. P1: SYN = 1

B. P2: SYN = 1, ACK = 1

C. P2: SYN = 0, ACK = 1

D. P3: SYN = 1, ACK = 1

[MSQ] (H.W.) IISC [GATE-2024][1 Mark]

#Q. TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

A. The sequence number N_P is always 0 for a new connection

B. The sequence number N_P is chosen randomly by P

C. The acknowledgement number N_Q is equal to N_P

D. The acknowledgement number N_Q is equal to $N_P + 1$

2 mins Summary

Topic TCP Header

Topic TCP Connections

THANK - YOU

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